

Desalegn Tamirat

Software Engineer

-  desalegntamirateshetu@gmail.com
-  +251979474672
-  Kenya, Nairobi
-  <https://www.linkedin.com/in/desalegn-tamirat-19727520a/>
-  <https://github.com/DesalegnTamirat>
-  <https://leetcode.com/u/DesalgenTamirat/>

PROFILE

Experienced Fullstack and Python Developer with extensive exposure to AI problem generation, LLM data evaluation, and software development. Strong background in Python, Django, React, Node.js, Express.js, TypeScript, and MongoDB. Proven experience designing high-throughput AI pipelines, mentoring and leading student teams, and building scalable fullstack applications. Passionate about AI research, problem-solving, and building efficient, reliable software systems.

LANGUAGES

Amharic ● ● ● ● ●

English ● ● ● ● ●

PROFESSIONAL EXPERIENCE

Head of Education — Instructor & Mentor

A2sv

06/2023 – Present

- Led **5 months of DSA training**, designing curriculum and problem sets (arrays, trees, graphs, DP), delivering live sessions, and mentoring students 1:1, which improved problem-solving consistency and interview readiness.
- Taught **fullstack web development (React, Node.js, Express, MongoDB, TypeScript)** and mentored 4 teams on projects like *Eskalate Job Agency* and *JobGen*.
- Guided students on **frontend design, backend REST API development, and database integration**, ensuring agile workflows, Git collaboration, and deployment best practices.
- Achieved deployment of functional fullstack applications for demos while reinforcing my own expertise in modern web development and team mentorship.

AI Problem Generation

Psynthetic

01/2024 – 08/2025

- Designed and implemented **pipelines for automated STEM problem generation** using Python and LLMs, ensuring logical structure and academic rigor.
- Improved **JSON parsing, LaTeX normalization, and error correction**, reducing malformed data and boosting pipeline reliability.
- Refined prompts, evaluated model outputs, and compared solution approaches to **enhance consistency and quality** of generated problems.
- Built Python scripts for **data extraction, logging, and error handling**, enabling high-throughput, large-scale dataset generation.
- Contributed to the production of **thousands of PhD-level STEM problems weekly**, strengthening AI evaluation reliability and advancing problem modeling techniques.

Training + Mentorship Program

A2sv

09/2021 – 06/2023

- Solved **1000+ problems** in algorithms, data structures, and system design.

SKILLS

Languages



Python, JavaScript, Typescript

Frameworks & Tools



Django, React, Express, Typescript, Nextjs

Database & Tools



MongoDB, SQL, Git, GitHub, VS Code

AI & ML



LLMs, PyTorch, TensorFlow, AI evaluation pipelines, problem modeling

Other



Data Structures & Algorithms, Problem Solving, Teaching & Mentorship, Critical Thinking, Analysis

- Collaborated in **peer coding sessions and mock interviews**, strengthening problem-solving and communication.
- Mentored juniors by breaking down complex concepts into approachable steps; one mentee successfully joined A2SV.
- Gained experience in **collaborative engineering workflows** using GitHub.

EDUCATION

Electrical and Computer Engineering

Addis Ababa University (AAiT)

2019 – 2024 | Addis Ababa, Ethiopia